

Srinivasan Poonkundran

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PROFESSIONAL EXPERIENCE

Research Assistant

Feb. 2026 – Present

The University of Arizona

Tucson, AZ

- Conduct data analysis and implement machine learning models to support ongoing research initiatives in collaboration with faculty and graduate researchers.
- Develop and evaluate computational workflows and experimental pipelines for data-driven studies.
- Contribute to documentation, reproducible research code, and preparation of technical materials.

Software Engineer

Sept. 2021 – Jan. 2024

Cognizant Technology Solutions

Remote

- Built Java Spring Boot microservices deployed on AWS, backed by MySQL databases, secured with OAuth2-based authentication and authorization, and integrated with Angular dashboards used by internal stakeholders.
- Improved API response time by 35% by optimizing SQL queries and caching critical endpoints.
- Built a Python-based PDF label validation tool to automate manual checks, reducing operational effort and saving approximately \$1.6K annually.
- Contributed to code reviews and refactored backend services with Java and Spring Boot, improving code quality and reducing defect occurrences, which accelerated release cycles.

Software Developer

Jan. 2023 – Jan. 2024

Self-Employed

Part-time, Remote

- Designed and deployed a production Android application supporting 1,000+ monthly transactions for a retail client.
- Built real-time billing, sales tracking, and revenue analytics dashboards within the mobile interface.
- Integrated AWS S3 for secure invoice storage, backup, and cross-device data retrieval.
- Implemented robust local storage and synchronization mechanisms to ensure data reliability under unstable network conditions.

PROJECTS

GradeLens – RAG-Powered AI Agent for Intelligent Evaluation | [GitHub](#)

- Architected a production-grade Retrieval-Augmented Generation (RAG) system using vector embeddings and LLM orchestration to enable scalable, context-aware automated grading.
- Designed modular ML pipelines for embedding generation, semantic retrieval, rubric validation, and structured output enforcement.
- Implemented asynchronous API handling with parallel execution to reduce grading latency across multiple exam questions.
- Deployed the system within a Django + React application, enabling scalable, low-latency inference.

Emotion Detection from Text | [GitHub](#)

- Designed and trained a transformer-based NLP classification system using BERT embeddings and Bi-GRU layers, achieving 0.83 F1-micro across 7 emotion classes.
- Built scalable preprocessing and training pipelines using Hugging Face and TensorFlow to efficiently handle large text datasets.
- Optimized model performance through structured tuning and regularization to improve generalization.

EDUCATION

The University of Arizona

Jan. 2024 – Dec. 2025

Master of Science, Data Science

Tucson, AZ

Honors: Distinguished Graduate Scholar (Fall 2025)

Savitribai Phule Pune University

Aug. 2017 – May 2021

Bachelor of Engineering, Computer Engineering

Pune, India

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, R, C, C++

Development Practices: Object-Oriented Programming, Data Structures & Algorithms, Microservice Architecture, Agile Methodologies, SDLC, CI/CD, Maven, Unit Testing (JUnit, Mockito)

Database & Cloud: MySQL, MongoDB, SQL/NoSQL, AWS - EC2, ECS, Load Balancers, Elastic Beanstalk, S3, IAM

AI Systems: Agno, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases (pgvector)

Certifications: Amazon Web Services Certified Cloud Practitioner | [Credentials](#)